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from google.colab import files

uploaded = files.upload()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")

print("Libraries Imported Successfully!")

df = pd.read_csv("retail_sales_dataset.csv")

print("\nDataset Loaded Successfully!")

print("\nFirst 5 Rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

df.fillna(method='ffill', inplace=True)

print("\nMissing Values Handled!")

duplicates = df.duplicated().sum()

print("\nDuplicate Rows:", duplicates)

df.drop_duplicates(inplace=True)

print("Duplicates Removed Successfully!")

df['Date'] = pd.to_datetime(df['Date'])

print("\nDate Column Converted Successfully!")

df['Month'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year

print("\nNew Features Created!")

print("\nDescriptive Statistics:")
print(df.describe())

print("\nMean Sales:", df['Total Amount'].mean())
print("Median Sales:", df['Total Amount'].median())
print("Mode Sales:", df['Total Amount'].mode()[0])
print("Standard Deviation:", df['Total Amount'].std())

total_sales = df['Total Amount'].sum()

print("\nTotal Sales:", total_sales)

gender_sales = df.groupby('Gender')['Total Amount'].sum()

print("\nSales by Gender:")
print(gender_sales)

plt.figure(figsize=(6,5))

gender_sales.plot(kind='bar', color=['skyblue', 'pink'])

plt.title("Sales by Gender")
plt.ylabel("Total Sales")

plt.show()
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plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=10, kde=True)

plt.title("Customer Age Distribution")

plt.show()

category_sales = df.groupby('Product Category')['Total Amount'].sum().sort_values(ascending=False)

print("\nSales by Product Category:")
print(category_sales)

plt.figure(figsize=(10,6))

sns.barplot(x=category_sales.index,
            y=category_sales.values)

plt.title("Product Category Sales")
plt.xlabel("Product Category")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.show()

monthly_sales = df.groupby('Month')['Total Amount'].sum()

print("\nMonthly Sales:")
print(monthly_sales)

# Plot
plt.figure(figsize=(10,5))

monthly_sales.plot(marker='o', linewidth=3)

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.grid(True)

plt.show()

top_customers = df.groupby('Customer ID')['Total Amount'].sum().sort_values(ascending=False).head(10)

print("\nTop 10 Customers:")
print(top_customers)

plt.figure(figsize=(10,5))

top_customers.plot(kind='bar')

plt.title("Top 10 Customers")
plt.ylabel("Sales")

plt.show()

plt.figure(figsize=(8,6))

sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()

plt.figure(figsize=(8,5))

sns.histplot(df['Total Amount'],
            bins=20,
            kde=True,
            color='green')

plt.title("Sales Distribution")

plt.show()

print("\n===== BUSINESS INSIGHTS =====")

best_category = category_sales.idxmax()
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print("\nTop Selling Category:", best_category)

best_month = monthly_sales.idxmax()

print("Best Sales Month:", best_month)

best_customer = top_customers.idxmax()

print("Top Customer ID:", best_customer)

print("\n===== RECOMMENDATIONS =====")

print("1. Increase stock for top-selling categories.")
print("2. Give special offers during low-sales months.")
print("3. Focus marketing on high-purchasing customers.")
print("4. Improve customer engagement programs.")
print("5. Use seasonal discounts to improve sales.")

df.to_csv("cleaned_retail_sales_data.csv", index=False)

print("\nCleaned Dataset Saved Successfully!")

print("\nEDA Project Completed Successfully!")
```


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Saving retail_sales_dataset.csv to retail_sales_dataset.csv

Libraries Imported Successfully!

Dataset Loaded Successfully!

First 5 Rows:

	Transaction ID	Date	Customer ID	Gender	Age	Product Category	\
0	1	2023-11-24	CUST001	Male	34	Beauty	
1	2	2023-02-27	CUST002	Female	26	Clothing	
2	3	2023-01-13	CUST003	Male	50	Electronics	
3	4	2023-05-21	CUST004	Male	37	Clothing	
4	5	2023-05-06	CUST005	Male	30	Beauty	

Quantity Price per Unit Total Amount

0	3	50	150
1	2	500	1000
2	1	30	30
3	1	500	500
4	2	50	100

Dataset Shape:

(1000, 9)

Column Names:

Index(['Transaction ID', 'Date', 'Customer ID', 'Gender', 'Age', 'Product Category', 'Quantity', 'Price per Unit', 'Total Amount'], dtype='object')

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1000 entries, 0 to 999

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Transaction ID	1000 non-null	int64
1	Date	1000 non-null	object
2	Customer ID	1000 non-null	object
3	Gender	1000 non-null	object
4	Age	1000 non-null	int64
5	Product Category	1000 non-null	object
6	Quantity	1000 non-null	int64
7	Price per Unit	1000 non-null	int64
8	Total Amount	1000 non-null	int64

dtypes: int64(5), object(4)

memory usage: 70.4+ KB

None

Missing Values:

Transaction ID 0

Date 0

Customer ID 0

Gender 0

Age 0

Product Category 0

Quantity 0

Price per Unit 0

Total Amount 0

dtype: int64

Missing Values Handled!

Duplicate Rows: 0

Duplicates Removed Successfully!

Date Column Converted Successfully!

New Features Created!

Descriptive Statistics:

	Transaction ID	Date	Age	Quantity	\
count	1000.000000	1000	1000.000000	1000.000000	
mean	500.500000	2023-07-03 00:25:55.200000	256	41.39200	2.514000
min	1.000000	2023-01-01 00:00:00	18.00000	1.000000	
25%	250.750000	2023-04-08 00:00:00	29.00000	1.000000	
50%	500.500000	2023-06-29 12:00:00	42.00000	3.000000	
75%	750.250000	2023-10-04 00:00:00	53.00000	4.000000	
max	1000.000000	2024-01-01 00:00:00	64.00000	4.000000	
std	288.819436	NaN	13.68143	1.132734	

Price per Unit Total Amount Month Year

count	1000.000000	1000.000000	1000.000000	1000.000000
mean	179.890000	456.000000	6.549000	2023.002000
min	25.000000	25.000000	1.000000	2023.000000
25%	30.000000	60.000000	4.000000	2023.000000
50%	50.000000	135.000000	6.000000	2023.000000
75%	300.000000	900.000000	10.000000	2023.000000
max	500.000000	2000.000000	12.000000	2024.000000
std	189.681356	559.997632	3.452755	0.044699

```
Mean Sales: 456.0  
Median Sales: 135.0  
Mode Sales: 50  
Standard Deviation: 559.997631555123
```

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Total Sales: 456000
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Sales by Gender:
```

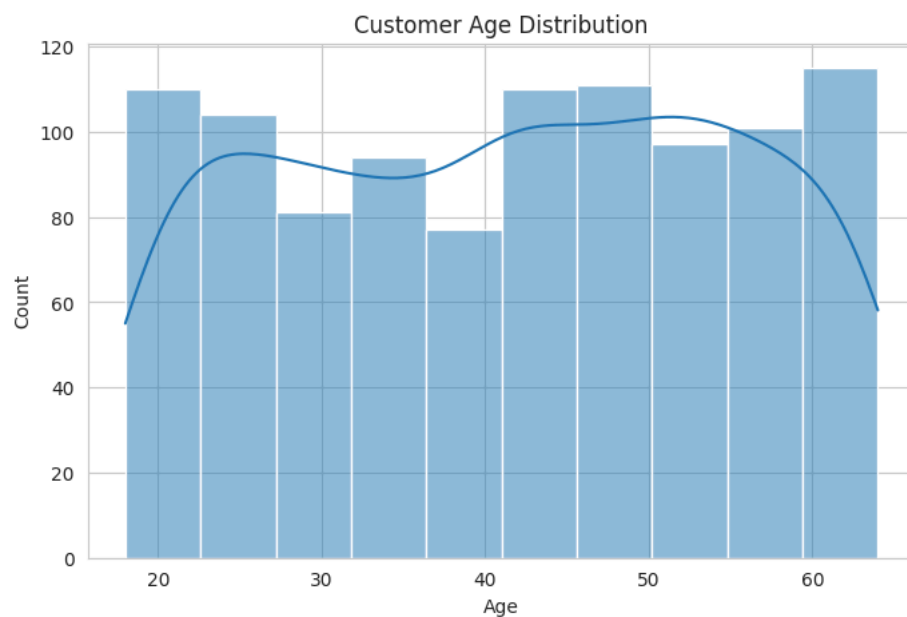
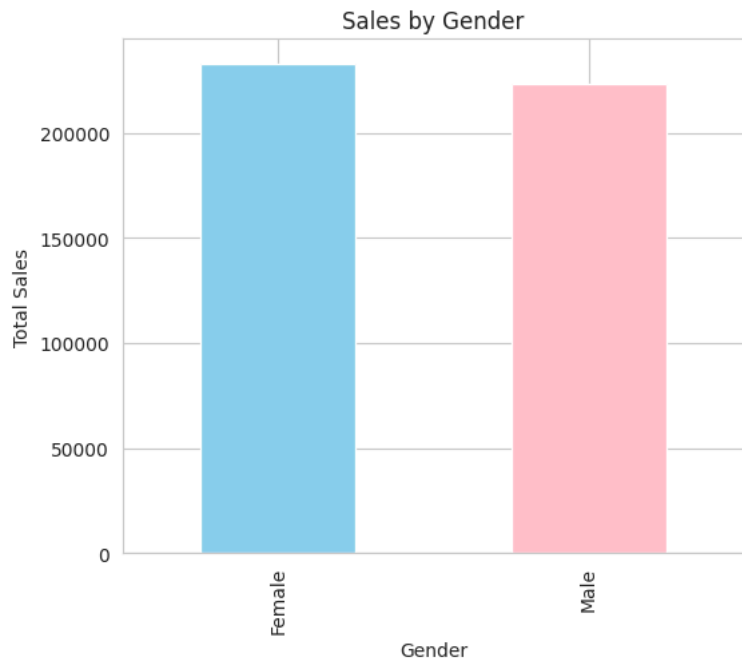
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Gender
```

```
Female    232840
```

```
Male      223160
```

```
Name: Total Amount, dtype: int64
```

```
/tmp/ipykernel_2779/1681394163.py:34: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use df.fillna(method='ffill', inplace=True)
```



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Sales by Product Category:
```

```
Product Category
```

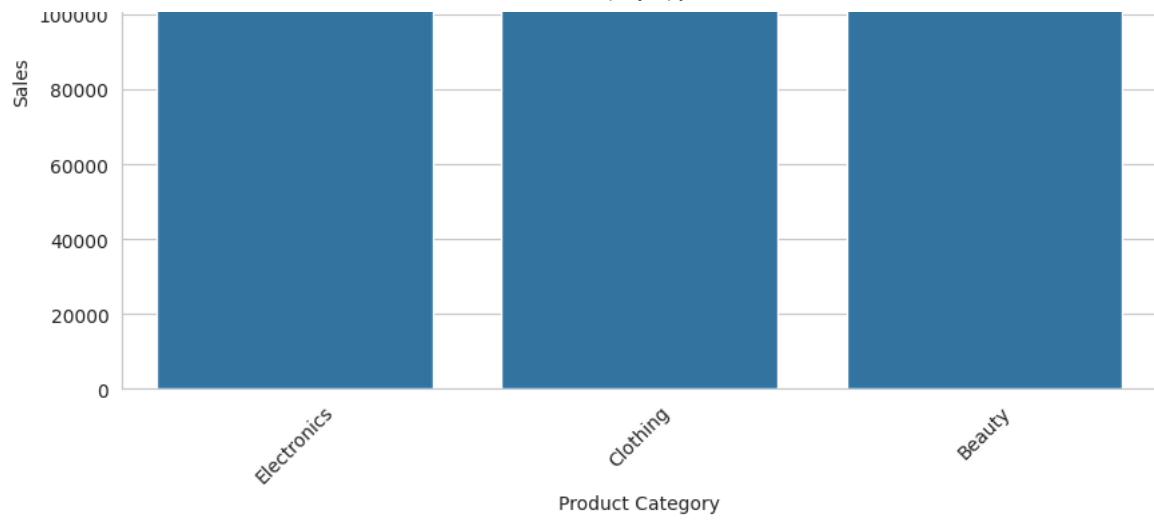
```
Electronics    156905
```

```
Clothing       155580
```

```
Beauty         143515
```

```
Name: Total Amount, dtype: int64
```





Monthly Sales:

Month

1	36980
2	44060
3	28990
4	33870
5	53150
6	36715
7	35465
8	36960
9	23620
10	46580
11	34920
12	44690

Name: Total Amount, dtype: int64

